EC39001: Analog Communication Lab Experiment 5: Frequency Modulation using XR-2206

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1 Aim of the Experiment

The objectives of this experiment are:

1. To realize **frequency modulation (FM)** using XR-2206 function generator IC.
2. To observe the FM signal in both the time domain and frequency domain using a Digital Storage Oscilloscope (DSO).
3. To determine the peak frequency deviation and modulation index.

2 Components Used

- XR-2206 Function Generator IC
- Breadboard
- Potentiometers: Frequency Control (R1), Offset Control (R5), Output Level Control (R6)
- Resistors and Capacitors (as per circuit diagram)
- Function Generator (for modulating signal)
- Mixed Signal Oscilloscope (DSO)
- $\pm 12V$ DC Power Supply
- Connecting Wires

3 Theory

3.1 Frequency Modulation (FM)

In frequency modulation, the instantaneous frequency of the carrier is varied according to the amplitude of the message signal. The general FM signal is represented as:

$$s(t) = A_c \cos\Big(2\pi f_c t + \beta \sin(2\pi f_m t)\Big)$$

where,

- A_c: Carrier amplitude
- f_c: Carrier frequency
- f_m: Message frequency
- $\beta = \frac{\Delta f}{f_m}$: Modulation index
- Δf: Peak frequency deviation

The frequency deviation Δf is given by:

$$\Delta f = f_{max} - f_c$$

where f_{max} is the maximum instantaneous frequency observed.

3.2 XR-2206 Function Generator IC

The XR-2206 is a monolithic function generator IC that can generate high-quality sine, square, and triangular waveforms. It consists of:

- A **Voltage-Controlled Oscillator (VCO)**.
- An **Analog Multiplier and Sine Shaper**.
- A **Unity Gain Buffer Amplifier**.
- A set of **Current Switches**.

For FM, the input modulating signal $m(t)$ is applied to the FM input pin, which controls the VCO. The output frequency is thus varied in proportion to the amplitude of $m(t)$, producing a frequency-modulated output.

4 Circuit Diagram

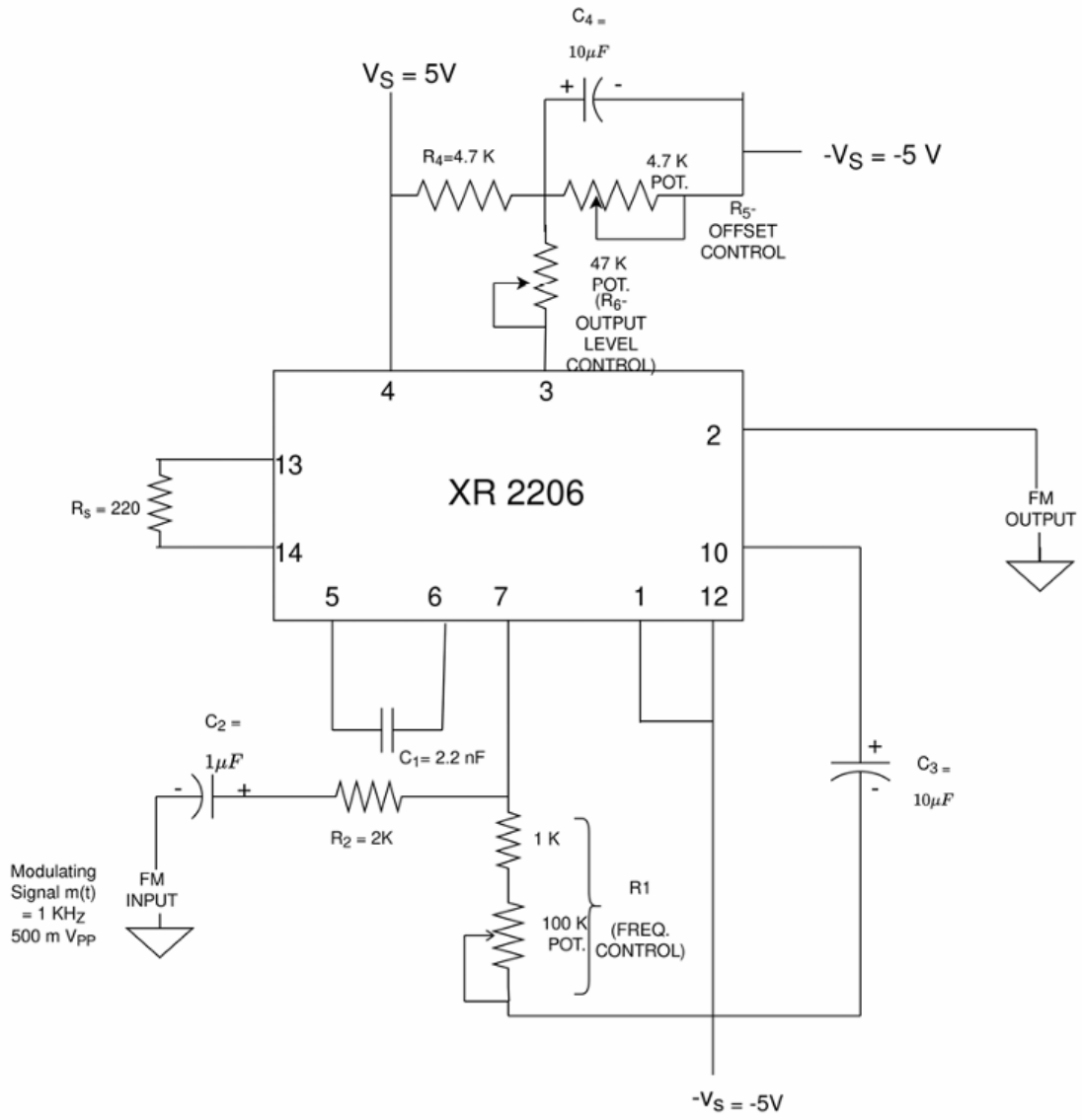


Figure 1: Circuit Diagram for Frequency Modulation using XR-2206

5 Observations and Results



Figure 2: Carrier Signal $f_c = 10\,kHz$

In Fig 2 we can see the clean high quality sine signal generated by XR 2206 , adjusted to $10\,kHz$ by setting appropiate value of R_1, R_5 and R_6.



Figure 3: Frequency Modulated Output Waveform with Modulating Signal

In Figure 3 we can observe modulated signal .The distinct feature of frequency modulation i.e change in the frequency of signal from high to low can easily be observed from the figure.



Figure 4: Frequency Spectrum of FM Signal with Message Signal Voltage 200mVpp

In Figure 4 we can observe the symmetric nature of the spectrum around the carrier frequency $f_c = 10\,kHz$ peaks occuring at $f_c \pm f_m$, $f_c \pm 2f_m$, $f_c \pm 3f_m$ etc ($10 \pm 1kHz, 10 \pm 2kHz, 10 \pm 3kHz$ etc), with highest peak occuring at $15\,kHz$ & $5\,kHz$.

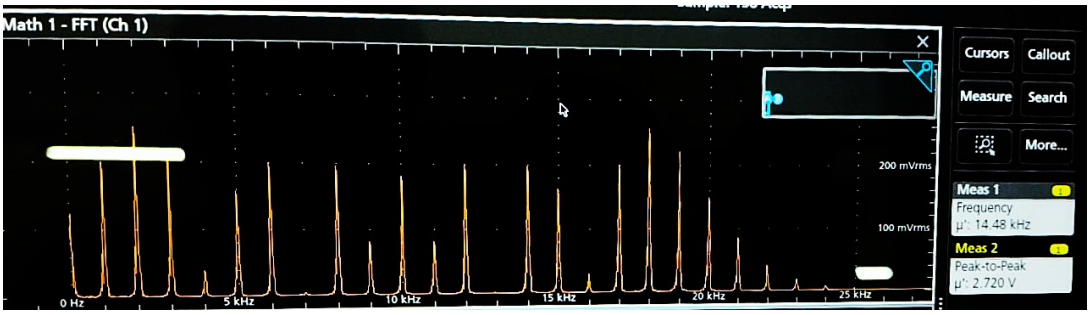


Figure 5: Frequency Spectrum of FM Signal with Message Signal Voltage 300mVpp

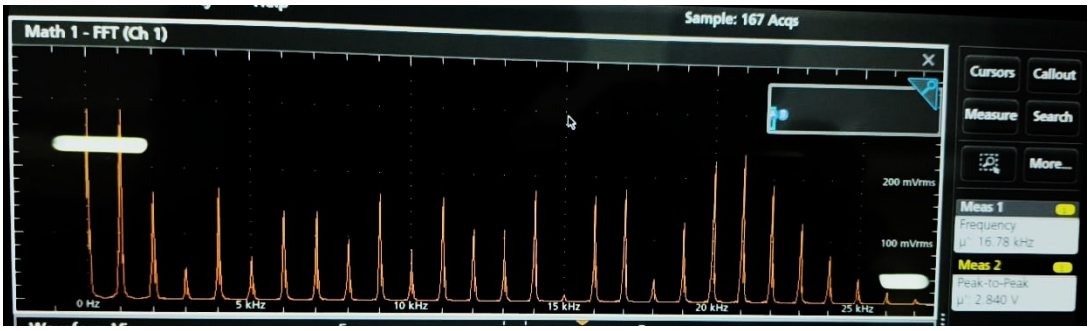


Figure 6: Frequency Spectrum of FM Signal with Message Signal Voltage 400mVpp

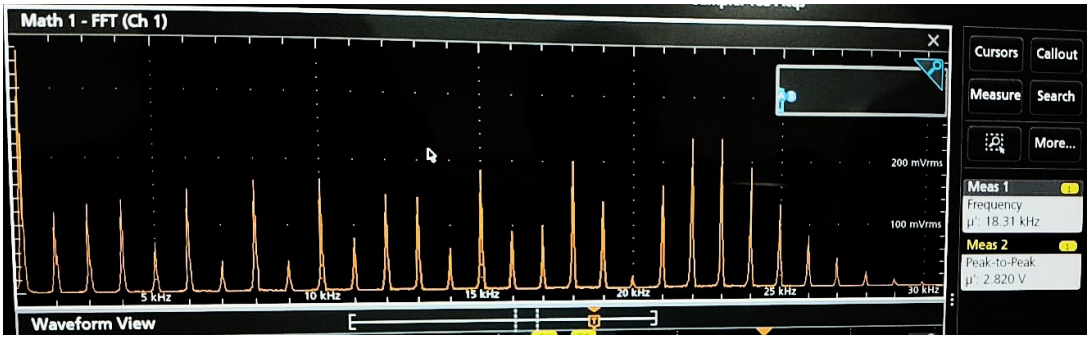


Figure 7: Frequency Spectrum of FM Signal with Message Signal Voltage 500mVpp

From Figure 4 ,5 ,6 and 7 ,We can observe that our frequency spectrum is spreading and the height of the peaks are changing . It can also be noted that the location of the highest peak is shifting away from f_c ,For Example - in Fig 4($A_m = 200mVpp$) highest peak was at $15\,kHz$,in Fig 5($A_m = 300mVpp$) highest peak was at $18\,kHz$,in Fig 6($A_m = 400mVpp$) highest peak was at $21\,kHz$ and for $A_m = 500mVpp$ highest peak was at $22kHz$ & $23kHz$.

From the observations:

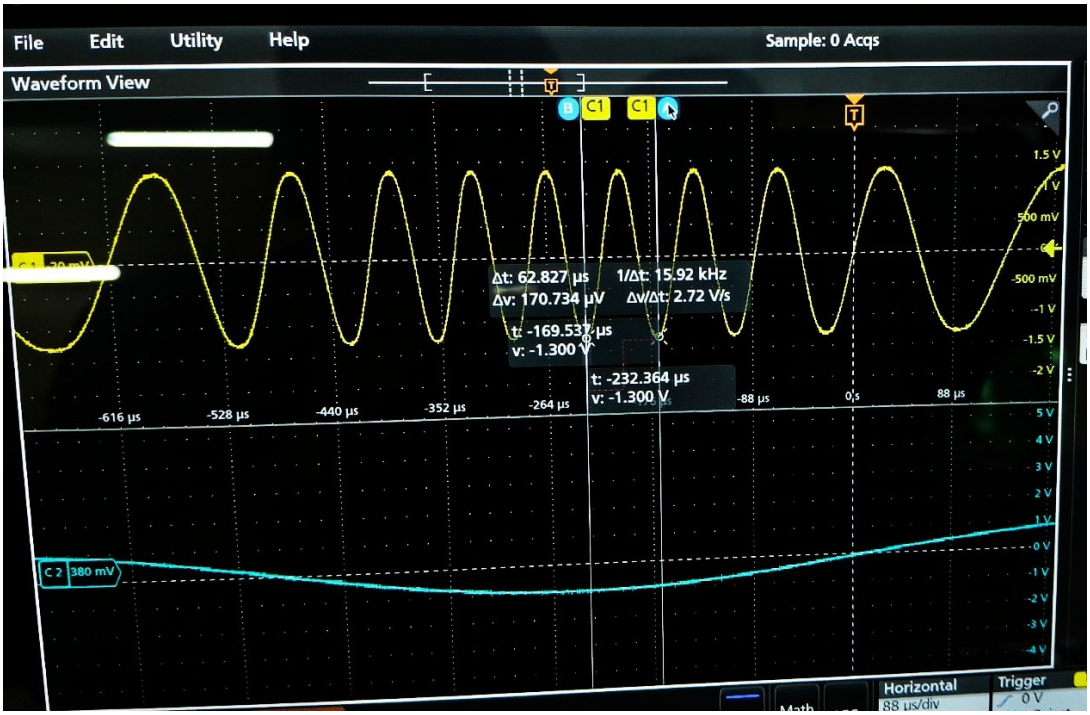


Figure 8: Measuring value of f_{max} and f_{min} from Output Waveform

- $f_c = 10\,kHz$
- $f_m = 1\,kHz$
- $f_{max} = 15.92\,kHz$ (from DSO)
- $f_{min} = 4.1\,kHz$ (from DSO)
- Peak Frequency Deviation: $\Delta f = f_{max} - f_c = 5.92\,kHz$
- Modulation Index: $\beta = \frac{\Delta f}{f_m} = 5.92$

6 Discussion

1. In this experiment we demonstrated frequency modulation using XR-2206 successfully.
2. Our first goal was to tune the XR 2206 chip to get the a clean high quality sine carrier wave .We tuned it using the different potentiometer to get a frequency of 10 kHz and peak to peak amplitude of 4 V, but due to high sensitivity of potentiometer

acheiving accurate value of 10 kHz and 4 V was difficult , we mangaed to tune it sufficiently close to acceptable value.

3. According to the manual the Amplitude value of message signal should be 500mVpp , but if we are using that value the frequency drift is so much that it is causing the phase to change its polarity suddenly causing distortion (similar to that caused by overmodulation). So we have used message signal amplitude to be 200mVpp, which is giving the expected outcome.

4. The Amplitude spectrum of the modulated signal for message amplitude of $200mVpp$ shows the symmetric distribution of sideband over the central carrier frequency ($f_c = 10\,kHz$).

5. We observe that as the modulating amplitude increases, the frequency spectrum becomes wider and the heights of the peaks gradually changing. Another noticeable trend is that the dominant peak shifts farther away from the carrier frequency f_c. For instance, in Fig.4 ($A_m = 200\,mV_{pp}$), the strongest peak appears at $15\,kHz$. In Fig.5 ($A_m = 300\,mV_{pp}$), it shifts to $18, kHz$, while in Fig. 6 ($A_m = 400, mV_{pp}$), the peak is located at $21, kHz$. Finally, for $A_m = 500\,mV_{pp}$, the major peaks are seen around $22\,kHz$ and $23\,kHz$.

6. This can be explained as the frequency width spectrum depends on the arguements of signal $s(t) = A_c \cos\Big(2\pi f_c t + \beta \sin(2\pi f_m t)\Big)$ which is $2\pi f_c t + \beta \sin(2\pi f_m t)$. As the value of message signal Amplitude increases the value of β increases causing increase in the spectral width or frequency deviation Δf of the signal ,confirming the FM principle.

7. Accurate tuning of XR-2206 controls was crucial for stabilizing carrier frequency and amplitude.And using right value of component was important to get noise free and undistorted signal.

——END OF REPORT——